# Hints: Functional sequences, series and spaces – 3. Power series

Homework

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

---

## Problem 1

**Hint 1.** Inside the smaller radius both series converge absolutely, and the Cauchy product of two absolutely convergent series converges to the product of the sums (Mertens; or rearrange the absolutely convergent double series). Grouping $a_k x^k \cdot b_m x^m$ by $k+m=n$ gives the power series with coefficients $c_n = \sum_{k=0}^{n} a_k b_{n-k}$.

**Hint 2.** For $\left(\sum \frac{x^n}{n!}\right)^2$: $c_n = \sum_k \frac{1}{k!(n-k)!} = \frac{1}{n!}\sum_k \binom{n}{k} = \frac{2^n}{n!}$. The classical divergent example is $\sum \frac{(-1)^n}{\sqrt{n+1}}$ multiplied by itself: the $n$-th term of the product has absolute value $\sum_k \frac{1}{\sqrt{(k+1)(n-k+1)}} \geqslant \frac{2(n+1)}{n+2}$, so it does not tend to 0.

## Problem 2

**Hint 1.** Look at $g(X) = Xf(X) = \sum_{n\geqslant 0} X^{2^n}$. In characteristic 2 squaring is additive, and $g(X)^2 + g(X) = X$; equivalently $Xf(X)^2 + f(X) = 1$.

**Hint 2.** Prove the identity first in the universal case $R_0 = \mathbb{F}_2[\alpha, \beta]$ (an integral domain, so $R_0[[X]]$ is too, and one may cancel), then specialize $\alpha, \beta$ by a ring homomorphism. Put $u = g(\alpha^2 X)$, $v = g(\beta^2 X)$, $w = g((\alpha+\beta)^2 X)$. Then $z = w - u - v$ satisfies $z^2 + z = 0$ with $z(0) = 0$; since $1 + z$ is invertible, $w = u + v$.

**Hint 3.** Let $A, B, C$ be the three factors, so $u = \alpha^2 XA$ etc. Using $u^2 = u + \alpha^2 X$ and $v^2 = v + \beta^2 X$: $uvw = u^2 v + uv^2 = X(\alpha^2 v + \beta^2 u) = \alpha^2\beta^2 X^2 (A + B)$, and $A + B = X(\alpha A + \beta B)^2$ from $A = 1 + \alpha^2 X A^2$. Hence $(\alpha+\beta)^2 ABC = (\alpha A + \beta B)^2$.

**Hint 4.** Finally, $\alpha A + \beta B$ is divisible by $\alpha + \beta$: if $f = \sum c_n X^n$, its coefficients are $c_n(\alpha^{2n+1} + \beta^{2n+1})$ and $\alpha^{2n+1} + \beta^{2n+1} = (\alpha+\beta)\sum_{i=0}^{2n} \alpha^i \beta^{2n-i}$. So $ABC = D^2$ with $D = \sum_n c_n \left(\sum_{i=0}^{2n} \alpha^i \beta^{2n-i}\right) X^n$, an identity with coefficients in $\mathbb{F}_2[\alpha, \beta]$ that specializes to any $R$.

## Problem 3

**Hint 1.** All three radii equal 1: the coefficients are $\geqslant 1$, and $d(n) \leqslant n$, $\sigma(n) \leqslant n^2$, while for partitions use $\sum p(n) x^n = \prod_{k\geqslant 1} \frac{1}{1-x^k}$, which converges for $0 \leqslant x < 1$.

**Hint 2.** Rewrite as Lambert-type series: $\sum d(n) x^n = \sum_k \frac{x^k}{1-x^k}$, $\sum \sigma(n) x^n = \sum_k \frac{k x^k}{1-x^k}$, $\ln \sum p(n) x^n = \sum_k -\ln(1-x^k) = \sum_m \frac{1}{m} \frac{x^m}{1-x^m}$. Then put $x = e^{-t}$, $t \sim 1 - x \to 0^+$, and compare the sums with Riemann sums/integrals.

**Hint 3.** Results: $\sum_k \frac{1}{e^{kt}-1} \sim \frac{1}{t}\ln\frac{1}{t}$; $\sum_k \frac{k}{e^{kt}-1} \sim \frac{1}{t^2}\int_0^\infty \frac{u\,du}{e^u - 1} = \frac{\pi^2}{6t^2}$; and $\ln \sum p(n) x^n \sim \sum_m \frac{1}{m^2 t} = \frac{\pi^2}{6t}$. So the growth orders are $\frac{1}{1-x}\ln\frac{1}{1-x}$, $\frac{\pi^2}{6(1-x)^2}$ and $\exp\left(\frac{\pi^2}{6(1-x)}(1 + o(1))\right)$.

## Problem 4

**Hint 1.** $\sum (-1)^n x^n = \frac{1}{1+x} \to \frac{1}{2}$ and $\sum_n (-1)^n n x^n = -\frac{x}{(1+x)^2} \to -\frac{1}{4}$.

**Hint 2.** Cesàro implies Abel: with $\sigma_n$ the Cesàro means, $\sum a_n x^n = (1-x)^2 \sum (n+1)\sigma_n x^n$ (Abel summation twice), and $(1-x)^2 \sum (n+1)x^n = 1$, so the Abel limit equals $\lim \sigma_n$ (this is Problem 3 of the class sheet). The series $\sum (-1)^n n$ is not Cesàro summable, because Cesàro summability forces $s_n/n \to 0$, while here $|s_n| \approx n/2$.

## Problem 5

**Hint 1.** If $\zeta^{2^m} = 1$, then $(r\zeta)^{2^n} = r^{2^n}$ for $n \geqslant m$, so $f(r\zeta) = \sum_{n<m} (r\zeta)^{2^n} + \sum_{n \geqslant m} r^{2^n}$, and the second sum tends to $+\infty$ as $r \to 1^-$ (monotone convergence).

**Hint 2.** Dyadic roots of unity are dense on the circle. An analytic continuation across an arc would extend $f$ continuously to a neighbourhood of the arc, so $f$ would be bounded near every point of the arc, contradicting the infinite radial limits at the dyadic roots in it.

## Problem 6

**Hint 1.** For $0 \leqslant x < 1$ and every $N$: $f(x) \geqslant \sum_{n \leqslant N} a_n x^n$, and the right-hand side tends to $\sum_{n \leqslant N} a_n$ as $x \to 1^-$. So $\liminf_{x \to 1^-} f(x) \geqslant \sum_{n \leqslant N} a_n$ for every $N$. (And $f$ is increasing, so the limit exists.)

**Hint 2.** Example: $a_n = \frac{1}{n^2}$; the radius is 1 and $0 \leqslant f \leqslant \frac{\pi^2}{6}$ on $[0, 1)$.

## Problem 7

**Hint 1.** Look for $g = \sum_{n \geqslant 1} b_n x^n$ with $f(g(x)) = x$. The coefficient of $x^n$ in $f \circ g$ is $a_1 b_n + P_n(b_1, \ldots, b_{n-1})$ for a polynomial $P_n$ (in the $a_k$), so the $b_n$ are determined recursively because $a_1 \neq 0$. A right inverse that is itself invertible is also a left inverse.

**Hint 2.** For $y = x + x^2$ solve the quadratic: $x = \frac{-1+\sqrt{1+4y}}{2} = y - y^2 + 2y^3 - \ldots$. Lagrange inversion: $[y^n]\, g = \frac{1}{n}[x^{n-1}]\left(\frac{x}{f(x)}\right)^n$; check it here with $\frac{x}{f(x)} = \frac{1}{1+x}$ (the coefficients are $(-1)^{n-1}\frac{1}{n}\binom{2n-2}{n-1}$).

## Problem 8

**Hint 1.** First the case of accumulation at the centre: if not all $a_n$ vanish and $a_k$ is the first non-zero one, then $f(x) = x^k(a_k + a_{k+1}x + \ldots)$, and the bracket is non-zero near 0, so the zeros of $f$ do not accumulate at 0. The same argument at any point $c$ uses the re-expansion of $f$ at $c$ (a power series is analytic inside its interval).

**Hint 2.** Let $S$ be the set of points of $(-R, R)$ where all derivatives of $f$ vanish. By the first hint an accumulation point of zeros lies in $S$, so $S \neq \varnothing$. $S$ is closed (continuity of the derivatives) and open (near a point of $S$ the re-expansion is identically 0). By connectedness $S = (-R, R)$, so $a_n = f^{(n)}(0)/n! = 0$. For the corollary, apply this to the difference of the two series.

## Problem 9

**Hint 1.** Fibonacci: the recurrence gives $F(x)(1 - x - x^2) = x$; decompose into partial fractions with the roots of $1 - x - x^2$ to get Binet's formula $F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$, $\varphi, \psi = \frac{1 \pm \sqrt{5}}{2}$.

**Hint 2.** Catalan: splitting at the first return gives $C(x) = 1 + xC(x)^2$, so $C(x) = \frac{1-\sqrt{1-4x}}{2x}$ (the sign is fixed by $C(0) = 1$); expanding $\sqrt{1-4x}$ binomially gives $c_n = \frac{1}{n+1}\binom{2n}{n}$. Euler: $\prod_k (1 + x^k) = \prod_k \frac{1-x^{2k}}{1-x^k} = \prod_k \frac{1}{1-x^{2k-1}}$, as all even factors cancel.

## Problem 10

**Hint 1.** If the radius exceeds 1, the series converges at $x = 1$ absolutely, so $a_n \to 0$; integers tending to 0 are eventually 0. Equivalently: a non-polynomial series with integer coefficients has infinitely many $|a_n| \geqslant 1$, hence radius at most 1.

**Hint 2.** Radius exactly 1 is possible for non-polynomials in both extreme ways: rational functions such as $\frac{1}{1-x}$, and series with a natural boundary such as $\sum x^{2^n}$. The Pólya–Carlson theorem says these are the only possibilities: a power series with integer coefficients and radius 1 is either a rational function or has the unit circle as a natural boundary.